Ionic Cheat Sheet

Ionic Framework 9.0.10 · Ionic CLI 7.2.1 · Capacitor 8.6.0 · Angular 22.2.5 · Node 22+ · AI-generated summary — verify against the official docs · see the Ionic pages for details

CLI — PROJECT

Install, scaffold, serve

```
npm install -g @ionic/cli
ionic start trailhead tabs --type=angular
ionic serve    # browser, live reload
```

Production web build

```
npx ng build --configuration production
```

Note

Angular 22 with standalone components and signals; example app id com.example.trailhead.

Capacitor 8 toolchain

```
Node.js 22+   Xcode 26.0+
Android Studio Otter (2025.2.1)+
iOS 15.0+     Android minSdk 24
compileSdk / targetSdk 36
```

PROJECT — LAYOUT

Paths

```
src/app/        components, services, routes
src/theme/variables.scss
                Ionic theme variables
src/global.scss global styles, Ionic CSS
android/, ios/  native projects, in Git
www/            web build output (webDir)
```

capacitor.config.ts

```
const config: CapacitorConfig = {
  appId: 'com.example.trailhead',
  appName: 'trailhead',
  webDir: 'www',
};
```

Note

webDir must match the folder the web build writes to (also holds server and plugins keys).

COMPONENTS — USE

Shell, header and body

```
ion-app, ion-router-outlet  # root, outlet
ion-header, ion-toolbar, ion-title
ion-content  # scrollable body, safe areas
```

Lists, input, actions

```
ion-list, ion-item, ion-label
ion-searchbar
ion-button, ion-fab, ion-icon
```

Tabs, overlays, paging

```
ion-tabs, ion-tab-bar, ion-tab-button
ion-modal, ion-alert, ion-toast
ion-refresher, ion-infinite-scroll
```

Standalone imports and icons

```
import { IonContent, IonHeader }
  from '@ionic/angular/standalone';
import { addIcons } from 'ionicons';
import { heart } from 'ionicons/icons';
// @Component({ imports: [IonHeader, ...] })
addIcons({ heart });
```

THEMING — STYLE

Global CSS variables

```
--ion-color-primary
  (+ -rgb, -contrast, -shade, -tint)
--ion-background-color
--ion-text-color
--ion-font-family
--ion-safe-area-top
```

Per-component custom properties

```
--background, --color
```

Force a mode (ios or md)

```
provideIonicAngular({ mode: 'md' })
```

Dark mode

Import one of Ionic's dark palette stylesheets or use prefers-color-scheme.

ROUTING — NAVIGATE

Routes (app.config.ts)

```
provideRouter(routes)
// lazy-load a page
loadComponent: () => import(...)
// guard
canActivate: [authGuard]   // CanActivateFn
```

Navigation

```
NavController   // programmatic
routerLink      // declarative
```

Page lifecycle

```
ionViewWillEnter  // about to show: refresh
ionViewDidEnter   // fully shown
ionViewWillLeave  // hiding
ionViewDidLeave   // hidden
ngOnInit  // once; cached pages skip it
```

CAPACITOR — SYNC

Native projects

```
npx cap add android    # or ios
npx cap open android   # or ios: IDE
npx cap run android    # device, emulator
```

Keep native in step

```
npx cap sync  # web, config, plugins
npx cap copy  # web build, config only
```

Maintenance

```
npx cap doctor   # env, plugin versions
npx cap migrate  # new Capacitor major
# icons and splash screens
npx capacitor-assets generate
```

PLUGINS — CALL

Camera 8.2.6 · Geolocation 8.2.4

```
Camera.getPhoto({
  resultType: CameraResultType.Uri })
Geolocation.getCurrentPosition()
Geolocation.checkPermissions()
Geolocation.requestPermissions()
```

Network 8.0.1 · App 8.1.2

```
Network.addListener(
  'networkStatusChange', cb)
App.addListener('appUrlOpen', cb)
```

Haptics 8.0.2 · Splash Screen 8.0.2

```
Haptics.impact()
SplashScreen.hide()
```

After installing

Run npx cap sync, declare permissions in AndroidManifest.xml and Info.plist, and wrap plugins in services so tests can replace them.

DATA — STORE

HttpClient with interceptors

```
provideHttpClient(withInterceptors([
  baseUrlInterceptor, authInterceptor]))
```

Native HTTP (capacitor.config.ts)

```
plugins: {
  CapacitorHttp: { enabled: true } }
```

Storage choices

```
settings  @capacitor/preferences 8.0.1
          Preferences.set, Preferences.get
files     @capacitor/filesystem 8.1.5
relational  @capacitor-community/sqlite
browser   IndexedDB
tokens    Keychain/Keystore-backed plugin
```

SHIP — TEST & RELEASE

Test and debug

```
npx ng test          # unit, component
npx playwright test  # e2e, PW 1.65.0
chrome://inspect     # Android WebView
Safari Develop menu  # iOS WebView
```

Android release

```
keytool -genkeypair -v \
  -keystore trailhead-upload.jks \
  -alias upload \
  -keyalg RSA -keysize 2048 -validity 10000
cd android
./gradlew assembleRelease  # release APK
./gradlew bundleRelease    # AAB, Play
```

iOS and versions

```
Xcode: Product, Archive  # TestFlight
versionCode / versionName  # Android
CFBundleVersion  # iOS build number
```

Over-the-air

@capacitor/live-updates 0.5.0 for optional web updates.